Exp No:18

Experiment to understand KNN algorithm for a given data set

AIM:

Understand the KNN algorithm for the dataset given.

PROGRAM:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Iris.csv')
df.info()
df.variety.value_counts()
df.head()
features=df.iloc[:,:-1].values
label=df.iloc[:,4].values
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.
2
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))
from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   sepal.length   150 non-null    float64
 1   sepal.width    150 non-null    float64
 2   petal.length   150 non-null    float64
 3   petal.width    150 non-null    float64
 4   variety        150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
Setosa         50
Versicolor     50
Virginica      50
Name: variety, dtype: int64
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
KNeighborsClassifier()
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

```
0.9583333333333334
1.0

array([[50,  0,  0],
       [ 0, 47,  3],
       [ 0,  2, 48]], dtype=int64)
```

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	50
Versicolor	0.96	0.94	0.95	50
Virginica	0.94	0.96	0.95	50
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150

RESULT:

The experiment has been carried out successfully